



Cover crop effects on X-ray computed tomography–derived soil pore characteristics

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Abstract

Purpose Cover crops have been used as an effective soil management practice to enhance soil health. However, this practice may create connected soil pore networks that can cause preferential transport of contaminants to the groundwater or surface water via subsurface flow pathways. The main objective of this study was to compare the effect of cover crops on the soil macropore characteristics in the soil profile.

Methods The study was conducted on soil columns collected from E.V. Smith Research Center, Shorter, AL, USA. This study evaluated the influence of cover crop (CC) vs. no cover crop (NC) on soil pore characteristics in strip-tillage cotton (*Gossypium hirsutum* L.). The cover crop treatment consisted of a mixture of cereal rye (*Secale cereale* L.) and crimson clover (*Trifolium incarnatum* L.). Six replicated intact undisturbed soil cores (150 mm diameter and 500 mm deep) were collected for the column study from each treatment class, i.e., CC and NC, and subjected to non-invasive X-ray computed tomography (CT) scanning, giving 0.35-mm-resolution images. The high-resolution images were analyzed in *ImageJ* to determine all the soil pore characteristics.

Results and discussion The results of the comparison of pore characteristics as a function of treatments showed that soil columns under CC had comparatively higher values of porosity and pore number density for the top 100 mm of soil. Pore geometry metrics such as tortuosity did not show significant differences among the treatments (CC vs NC). Connection probability was significantly higher for CC in the subsurface depth class (200–400 mm). Significant correlations were also observed between CT-derived pore characteristics and root characteristics from which it can be inferred that cover crop roots influenced the X-ray CT-derived pore properties.

Conclusions Cover cropping significantly impacted the macropore properties of the strip-till cotton field. This was attributed primarily to the influence of root networks on macropores. Our study's correlations between root properties and macropore characteristics also indicated that larger root volumes were significantly correlated with complex and irregularly shaped macropores. These variables are critical for a better understanding of the flow dynamics of contaminants through the soil profile and for developing appropriate management strategies.

Keywords Cover crops · Roots networks · Macropores · X-ray CT · Vadose zone

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1 Introduction

Conservation tillage and cover crops have been shown to improve soil and water quality and crop yield (Munkholm et al. 2013). These conservation agriculture practices can also affect soil physical properties, such as the number of pores and pore size distribution of the soil profile over time (Wolschick et al. 2021; Kumar et al. 2021) which can further influence nutrient leaching (Miranda-Vélez et al. 2023) through preferential flow pathways in agricultural soils

(Filipović et al. 2020). The presence of these rapid flow pathways reduces the contact time of contaminants within the soil matrix and can therefore result in the loss of pollutants from agricultural landscapes (Qiu et al. 2013).

Cover crops are known to enhance soil health by reducing soil erosion and surface runoff and improving infiltration (Kaspar and Singer 2011) because of aerating the surface soil (Haruna et al. 2020) and improving soil organic carbon contents (Çerçioğlu 2020). Winter cover crops such as cereal rye and crimson clover are commonly planted in the southeastern United States to improve soil properties (Kornecki et al. 2016). Previous studies have reported that cover crops such as cereal rye and crimson clover can reduce soil loss and runoff, decrease bulk density, and enhance porosity and saturated hydraulic conductivity (Kornecki 2018). Although cover crops have been shown to improve the overall soil health, they can create continuous pores which can facilitate the transport of contaminants via preferential flow pathways through soil macropores (Saravanathiiban et al. 2014), especially in areas where subsurface/lateral flows are significant.

Apart from cover crop systems, conservation tillage practices improves soil properties such as soil structure, bulk density, and water holding capacity; thereby, leading to enhanced crop productivity (Busari et al. 2015; Ketema and Yimer 2014). On the other hand, conservation tillage practices have been shown to contribute towards the formation of massive vertical pores (Garbout et al. 2013), or well-connected pore networks (Dal Ferro et al. 2014) which can cause contaminant transport through the soil profile. Therefore, it is important to investigate the role that conservation tillage has on soil macropore properties which will ultimately influence contaminant transport and dynamics in the subsurface environment.

Contaminant transport pathways along the porous media of a soil profile can be traced using various mechanisms (Anderson et al. 2015). Previous studies have been conducted to visualize these soil pore pathways using different methods. For instance, dye tests were used to trace the macropore patterns (Noguchi et al. 1999; Alaoui and Goetz 2008; Wang and Zhang 2011), but it is known to be a destructive method and can lead to overestimation of macropore pathways as the dye can be absorbed into the pore walls (Filipović et al. 2020). Tension infiltrometers have been used to detect hydrologically active macropores at a given tension using in situ infiltration measurements (Buttle and McDonald 2002; Müller et al. 2018). Macropore quantification has also been done using microscopic analysis on thin sections of soil (Liu et al. 2016), also known as image analysis of impregnated soil blocks (Schlüter et al. 2019). This method is labor-intensive and gives only two-dimensional visualization (Taina et al. 2008; Li and Shao 2020). X-ray computed tomography (CT) is a non-destructive means to quantify the three-dimensional visualization of the

spatial distribution of soil pore networks (Hu et al. 2019; Zhang et al. 2017; Li et al. 2018). Out of all the above mentioned methods, X-ray CT is the most robust technique (Rabot et al. 2018) to determine pore shape (Saravanathiiban et al. 2014; Yang et al. 2021), precise estimation of pore size distribution and soil structure complexity (Muñoz et al. 2014), pore connectivity (Renard and Allard 2013; Larsbo et al. 2014; Lucas et al. 2021; Bölscher et al. 2021), pore inter-connection, and isotropy via non-destructive means (Hellner et al. 2018). The information on soil properties derived from X-ray CT scanning can be thereafter used to link the pore size distribution and pore connectivity metrics to the pore network simulation models (Tracy et al. 2015), solute transport (Wildenschild et al. 2002), and greenhouse gas emissions (Mangalassery et al. 2014; Mao et al. 2020) which further impact the climate change processes (Kochiieru et al. 2018; Budhathoki et al. 2022a).

X-ray CT scanning has been used in the past to evaluate pore characteristics. However, most previous studies quantifying the effects of cover crops on macropore networks in row-cropping systems using X-ray CT scanning have focused on corn (*Zea mays* L.) and soybean (*Glycine max* (L.) Merr.) cropping systems (Cercioglu et al. 2018; Singh et al. 2020; Dhaliwal and Kumar 2022; Rankoth et al. 2022). Cotton is an important crop grown in various parts of the world, including the southeastern United States. Understanding the size, shape, and connectivity of macropores in cotton fields is of paramount importance, as it can provide valuable information about macropore networks beneath cotton fields and aid in the development of more sustainable cotton production systems. Furthermore, as cover cropping systems are increasingly adopted as a conservation practice, more research is required to elucidate soil macropore characteristics in these systems. This is critically important in areas where contaminant loss can be substantial via lateral or subsurface flow pathways. To date, research studies quantifying the effects of cover crop systems on CT-derived pore network properties are scarce, particularly in landscapes planted with cotton. As a result, it is important to quantify soil macropore properties in strip-tillage systems of fields planted with cotton as the main crop following a cover crop. Hence, the main objectives of this article were as follows: (1) to compare the effects of cover crop treatment (CC) with no-cover crop treatment (NC) on X-ray CT-derived soil pore networks, (2) to correlate the X-ray CT-derived soil pore characteristics with root properties. These research results will provide valuable data on soil macropore characteristics which will help develop decision support tools. The dataset generated from this study alongside other published literature on soil macropore characteristics in corn and soybean cropping systems will aid in the development of management practices that can help mitigate nutrient losses in lateral or subsurface flow pathways.

2 Materials and methods

2.1 Experimental site and treatments

The study was conducted using a compass series soil (coarse-loamy, siliceous, subactive, thermic *Plinthic Paleudults*) collected from the Alabama Agricultural Experiment Station's E.V. Smith Research and Extension Center (32°26'25.9"N, 85°53'26.2"W) near Shorter, Alabama. The Compass series is representative of a Coastal Plain soil. These soils are mainly well-drained loamy sands. The horizon designation according to the United States Department of Agriculture (USDA) soil taxonomy classes was Ap and loamy sand for the top 300 mm, thereafter BE and sandy loam for 300–400 mm, and Bt and sandy loam 400–500 mm of the soil profile. The treatments consisted of comparing cover crop vs. no cover crop in strip-tillage cotton (*Gossypium hirsutum* L.). For the cover crop treatment, a mixture of 33.6 kg ha⁻¹ of cereal rye (*Secale cereale* L.) seed and 22.4 kg ha⁻¹ of crimson clover (*Trifolium incarnatum* L.) seed was planted to establish the cover crop. The cover crop treatments were initiated in the fall of 2019 prior to planting cotton as the main crop in 2020. The cotton was seeded at 134,500 seeds ha⁻¹.

2.2 Soil sampling

A total of 12 intact undisturbed soil cores for the column study (six replicates each from the CC and NC treatments) were collected after harvesting the cotton in 2021. The soil cores were collected using Schedule 40 PVC pipes with a 150 mm inside diameter and 640 mm height. These pipes were pushed vertically downward into the soil using a tractor-mounted hydraulic system (Prior et al. 2004) and excavated carefully with a backhoe and hand tools. The depth of soil in each PVC pipe was 500 mm with soil, leaving the top 140 mm of pipe containing air, similar to that described by Koestel et al. (2021). After excavation, the bottom of each soil column was trimmed using a sharp-edged steel plate to attain a horizontal planar soil surface, flush with the bottom of the pipe. Afterward, we placed bubble wrapping material against the soil surface at the upper and lower ends of the column. End caps were used to securely seal the columns to prevent disturbance during handling and transportation. All the soil columns were stored at 4 °C prior to any further analysis to prevent the growth of plants and organisms (Terribile et al. 2022). In the laboratory, the columns were saturated by capillary saturation and allowed to drain for 3 days to ensure soil moisture consistency. This procedure was performed to normalize the X-ray attenuation value for all the soil samples (Luo et al. 2010b).

Additional soil samples were collected using an auger (50 mm diameter and 100 mm deep) for measurement of soil properties such as organic matter content (%), particle size distribution (%), pH, electrical conductivity (EC) (dS m⁻¹), bulk density (g cm⁻³), and saturated hydraulic conductivity (cm day⁻¹) under each treatment (two replicates each for CC and NC treatment) at five depth classes (0–100, 100–200 mm, 200–300 mm, 300–400 mm, and 400–500 mm). The organic matter content was determined using the loss on ignition method (Schulte and Hopkins 1996). Particle size distribution was determined using the hydrometer method (Bouyoucos 1962), which gave the percentages of sand, silt, and clay in the soil samples at various depths. The values of pH and EC were measured using pH and EC meters, respectively, on soil slurries made from air-dried soil and distilled water (Zhang and Wienhold 2002). Bulk density was calculated by dividing the soil's oven-dried mass by the soil's volume. The saturated hydraulic conductivity of soil samples was determined using the Rosetta pedo transfer function model (Schaap et al. 2001) from the Rosetta-soil 0.0.1 (Rosetta 3) python package (Zhang and Schaap 2017). This function can be used to predict the value of saturated hydraulic conductivity using the values of bulk density and sand, silt, and clay percentages of the soil samples. All the values of the measured soil properties are reported in Table 1.

2.3 X-ray CT scanning of soil samples

All the soil samples were CT scanned using a GE Light-Speed VCT 64 Slice medical CT scanner (GE Healthcare, Chicago, IL) installed at the Auburn University College of Veterinary Medicine. The X-ray scans were carried out at a voltage of 140 kV and a current of 140 mA. The exposure time was 1 s, and the slice thickness was 0.625 mm. The field of view was set to 180 mm with a matrix size of 512×512 pixels which gave 16-bit DICOM images of 0.35 mm resolution.

2.4 Image acquisition, processing, and analysis

All the image processing was done using the FIJI distribution of *ImageJ* 2.3.0/1.52q image processing software (Schindelin et al. 2012). The DICOM 16-bit grayscale images received from X-ray CT scanning were combined to form Tiff (tagged image file format) image stacks of 0.35×0.35×0.625 mm voxel size. The images were aligned using the template matching plugin of FIJI. The gray value histogram was used to set the range of brightness and contrast of the images. Images were preprocessed using a non-linear rank median filter to maintain better edges for thresholding and measurement for a defined region of interest (ROI). An unsharp mask was used preceding the median

Table 1 Mean ($n=2$) values of soil properties under cover crop (CC) and no cover crop (NC) treatments for various depth classes. Values shown in parenthesis are standard deviations

Depth (mm)	Treatment	Organic matter (%)	Particle size distribution (%)			pH	Electrical conductivity (dS m ⁻¹)	Bulk density (g cm ⁻³)	Saturated hydraulic conductivity (cm day ⁻¹)
			Sand (0.05–2.00 mm)	Silt (0.002–0.05 mm)	Clay (< 0.002 mm)				
0–100	CC	1.46 (±0.01)	84.96 (±0.06)	9.35 (±1.01)	5.69 (±1.06)	5.07 (±0.30)	118.62 (±6.30)	1.30 (±0.06)	290.75 (±11.11)
	NC	0.90 (±0.29)	85.78 (±0.68)	10.54 (±0.68)	3.68 (±0.00)	5.00 (±0.12)	139.20 (±6.43)	1.49 (±0.16)	244.80 (±107.38)
100–200	CC	1.70 (±0.15)	84.79 (±1.11)	8.89 (±3.78)	6.32 (±2.68)	5.42 (±0.39)	86.70 (±17.14)	1.41 (±0.03)	228.72 (±17.66)
	NC	1.18 (±0.00)	85.24 (±0.26)	10.92 (±0.54)	3.84 (±0.27)	5.45 (±0.03)	100.95 (±0.49)	1.63 (±0.05)	153.22 (±28.99)
200–300	CC	1.31 (±0.01)	85.78 (±0.30)	9.04 (±3.99)	5.18 (±4.29)	5.37 (±0.78)	89.77 (±53.46)	1.50 (±0.02)	222.53 (±47.27)
	NC	1.28 (±0.16)	85.24 (±0.26)	10.92 (±0.54)	3.84 (±0.27)	5.49 (±0.66)	78.65 (±15.49)	1.66 (±0.08)	136.68 (±41.42)
300–400	CC	1.31 (±0.32)	78.18 (±0.25)	13.58 (±0.42)	8.24 (±0.17)	5.79 (±0.20)	65.52 (±3.68)	1.47 (±0.04)	101.12 (±9.91)
	NC	1.23 (±0.01)	77.00 (±1.41)	13.84 (±0.12)	9.16 (±1.30)	5.32 (±0.21)	125.57 (±0.28)	1.40 (±0.09)	111.76 (±42.54)
400–500	CC	1.42 (±0.00)	78.71 (±1.01)	12.50 (±1.94)	8.78 (±0.94)	5.10 (±0.18)	134.15 (±11.67)	1.79 (±0.01)	27.50 (±1.72)
	NC	1.57 (±0.18)	78.50 (±2.12)	12.40 (±3.57)	9.10 (±1.45)	4.53 (±1.38)	305.70 (±123.18)	1.87 (±0.19)	21.99 (±13.48)

filter for further edge enhancement. Images were then converted to 8-bit format for thresholding.

Segmentation is the most important step in image analysis as it binarizes the whole image into 2 values, i.e., 0 and 255, and is based on the principle of differentiating the background pixels from the foreground pixels (Li and Shao 2020). Auto-local thresholding was chosen as it is known to provide accurate results compared to global thresholding, which might cause misclassification errors (Taina et al. 2008). The best algorithm was chosen by visual inspection and by the “try all” in the auto local thresholding plugin. The Phansalkar method was chosen as it gave the best segmentation out of all the algorithms (Phansalkar et al. 2011). Most of the studies related to X-ray CT scanning of agricultural soils have found this method suitable for image segmentation (Müller et al. 2018; Pulido-Moncada et al. 2020; Singh et al. 2021; Budhathoki et al. 2022a; Chakraborty et al. 2022). To specify the values of the radius window, Parameter 1 and Parameter 2 for the Phansalkar method, Plexiglas thermoplastic cylindrical rods of known diameter (1.59 mm, 3.17 mm, and 4.76 mm) were inserted and removed before CT scanning in two soil columns to create artificial pores. Various values of the thresholding radius (which ranges from 0 to 15), Parameter 1, i.e., k (ranges from 0.2 to 0.5), and Parameter 2, i.e., r (has a value of 0.5 for a normalized image), were tested until the image derived diameter

of the artificially created pore was within 1.5% of the actual diameter of the rod (Budhathoki et al. 2022b). Chu et al. (2020) did a research study to determine various values of the parameters of Phansalkar auto local thresholding and reported that a smaller thresholding radius gave better binarization results (in our case thresholding radius was chosen as, 5). This value of radius has shown to cover both the background and the foreground (Nichele et al. 2020). Parameter 1, i.e., k , was used to control the value of threshold radius (Shafait et al. 2008), and it gave the best results at a value of 0.35, and Parameter 2 gave the best results at the default value of 0.5.

The pore voxels were made isotropic for further three-dimensional analyses. The pore size, pore shape, and pore connectivity analyses were conducted using plugins available in FIJI, i.e., analyze particles, SoilJ (Koestel 2018), and BoneJ (Doubé et al. 2010). These plugins gave the basic pore statistics, such as the pore area, pore number, and pore volume, from which all the pore characteristics were derived. The pore volume was derived from the number of pore voxels of each pore multiplied by the voxel volume calculated from the image resolution (0.35 * 0.35 * 0.35 mm) (Peth et al. 2008). Due to resolution limitations and partial volume effects, we have considered all the pores with a diameter of more than 0.7 mm as macropores (Budhathoki et al. 2022a, c). All the properties calculated from the CT-derived

values were classified into three main subcategories, i.e., pore volume descriptors, pore geometry descriptors, and pore connectivity descriptors. The definitions for the volume descriptors such as porosity (Luo et al. 2010a), pore number density (Katuwal et al. 2015), surface area density (Budhathoki et al. 2022a), and the geometry descriptors such as macropore diameter using local thickness algorithm in BoneJ (Gantzer and Anderson 2002; Larsbo et al. 2014; Zhang et al. 2018), tortuosity (Saravanathiiban et al. 2014), and circularity (Yang et al. 2020), and pore connectivity descriptors such as interconnectivity (Luo et al. 2010b; Munkholm et al. 2013), connection probability (Renard and Allard 2013; Jarvis et al. 2017), and Euler number density (Lucas et al. 2021) are summarized in Table 2.

These macropore properties derived from X-ray CT scanning were used to test correlations among the pore networks and root networks present within the soil. For root properties quantification, roots were obtained from the soil cores by manually extracting the root samples from the soil columns at 100 mm depth increments. After separating the soil chunks into various depth classes, soil samples were washed and sieved to extract the roots from the samples. These roots were scanned using a WinRhizo Pro 2020a scanner (Regents Instruments, Quebec City, QC, Canada). Root characteristics such as root length, mean root diameter, root volume, and root area were derived from the scanned images (Hu et al. 2019).

3 Statistical analysis

All the X-ray CT properties are expressed as means of six replicates per treatment. The normality of different variables was tested by the Shapiro–Wilk test, and the homogeneity of variances was checked using Levene’s test. Even after applying various transformations, the X-ray CT-derived soil pore parameters were not normally distributed, so a non-parametric Kruskal–Wallis test was used to detect significant differences at a probability value of 0.05 between different treatments (CC vs NC) at different depths of the soil profile using SAS 9.4 (SAS Institute, Cary, NC) and Jupyter Lab 3.0.14. Spearman’s correlation analysis was used to define relationships among root characteristics and CT-derived pore properties such as porosity, pore number density, and circularity.

4 Results and discussion

4.1 3D visualization of pores along the soil profile

Three-dimensional images of representative cores obtained after image processing from the two treatments are shown in Fig. 1. Based on visual observations, pores along the soil profile tended to be more concentrated as depth increased

for cores of the CC treatment. Cores from the CC system had a high number of pores as compared to the NC system. Çerçioğlu (2020) suggested soil with cover crops has a greater number of pores (both macro and micropores) because of the more extensive root networks and more organic matter. Another possible reason mentioned by Rankoth et al. (2022) was that increased microbial activity from cover crops led to the greater formation of biopores. These images of representative soil cores from our study show a dense network of complex connected macropores, which can act as a conduit for contaminant transport in the subsurface environment.

4.2 Pore volume descriptors

Five depth increments (0–100 mm, 100–200 mm, 200–300 mm, 300–400 mm, and 400–500 mm), two treatment classes (CC and NC), and five volumetric pore size groups ($< 1 \text{ mm}^3$, $1\text{--}10 \text{ mm}^3$, $10\text{--}100 \text{ mm}^3$, $100\text{--}1000 \text{ mm}^3$, and $> 1000 \text{ mm}^3$) were used to explain the variation among the various CT measured pore characteristics. The variation of porosity values along the soil profile for various volumetric pore size groups is shown in Fig. 2. Porosity varied significantly ($p < 0.05$) among the different treatments for all four volumetric size subclasses. The CC system had significantly ($p < 0.05$) greater porosity for all volumetric pore size classes at the surface layer (0–100 mm). This could be attributed to a greater density of pores formed by roots in the CC treatment (Çerçioğlu et al. 2019), leading to greater porosity values in this depth zone. The NC treatment system had the lowest porosity value for all size classes throughout the soil profile at different depths except for pores $> 1000 \text{ mm}^3$ at the 100–200 mm depth. Helliwell et al. (2019) and Koebernick et al. (2019) indicated that although plant roots may cause soil densification, they can create high porosity zones within the soil profile. However, Phalempin et al. (2021) reported that roots did not increase the overall porosity values in a maize field experiment, as roots grew into existing pore space. At the 100–200 mm depth, soil columns from the CC treatments had significantly ($p < 0.05$) lower porosity compared to NC. This can be explained by the fact that cover crop roots only went down to approximately 250 mm; therefore, porosity formation was possibly limited by depth. Haruna et al. (2018) and Budhathoki et al. (2022a) had similar results showing that reduced porosity as depth increased was linked to the formation of a hard pan or plow pan under a corn-soybean system and a pasture land, respectively. There were no significant differences among porosity values for the bottommost layer of the soil core (400–500 mm) for all volumetric size classes except $> 1000 \text{ mm}^3$. For the CC treatment, pores $> 1000 \text{ mm}^3$ were responsible for a majority of the total porosity. Smaller-sized pores had a lower contribution to the overall total porosity

Table 2 Soil pore characteristics derived from X-ray CT scanned images

S. no.	Pore characteristics	Alias	Definition	Formula	Range/interpretation	Units
1	Pore volume descriptors					
	Macroporosity	η	Reflection of the spatial arrangement of mineral particles and aggregates, as well as biological activity	$\frac{\text{Total volume of macropores}}{\text{Volume of ROI}}$	Greater macroporosity relates to the fact that more macropores interconnect to form bigger macropores	$\frac{\text{mm}^3}{\text{mm}^3}$
	Macropore number density	MND	Number of macropores per unit volume	$\frac{\text{Number of macropores}}{\text{Volume of ROI}}$	Greater number of isolated macropores lead to better aeration of soil whereas greater number of connected macropores can lead to rapid leaching of contaminants	$\frac{\text{Number}}{\text{mm}^3}$
2	Pore geometry descriptors					
	Surface area Density	σ	Specific surface area of macropore	$\frac{\text{Surface area of macropores}}{\text{Volume of ROI}}$	Greater values of specific surface area are associated with irregular pores causing greater resistance to flow	$\frac{\text{mm}^2}{\text{mm}^3}$
	Macropore diameter	d	It assumes the pore area to be circular in 2D and inscribes a sphere of maximum diameter in 3D	$\sqrt{\frac{2 \times \text{pore area}}{\pi}}$	< 1 mm very fine macropores; 1–2 mm fine macropores; 2–5 mm medium macropores; > 5 mm coarse macropores	mm
	Circularity	C	Represents the characteristics of pore morphology	$\frac{4 \times \pi \times \text{pore area}}{\text{pore perimeter}}$	0–1, closer to 1 more circular	N/A
	Tortuosity	τ	Relates straight line distances between centroids of pores to their actual lengths	$\frac{\text{Actual macropore length}}{\text{Straight line distance}}$	Highly continuous macropores have relatively low tortuosity	N/A
3	Pore connectivity descriptors					
	Interconnectivity	I	Intersections of at least 2 pore branches of macropore network	$\frac{\sum \text{number of nodes}}{\text{Volume of ROI}}$	High density corresponds to a well-connected network	
	Connection Probability	Γ	Probability of 2 pore voxels to be a part of the same macropore cluster	$\frac{1}{N_p^2} * \sum_{k=1}^{N_i} n_k^2$ N_p = number of pore voxels N_i = number of all clusters n_k = number of pore voxels in cluster k	Ranges from 0 to 1, 1 when all pores are from same cluster; higher if large fraction of porosity belongs to large connected cluster	N/A
	Euler Number	χ	Number of independent paths between 2 boundaries	Number of unconnected clusters – total number of redundant loops + total number of closed cavities	Negative; if number of redundant connections > number of unconnected clusters	N/A

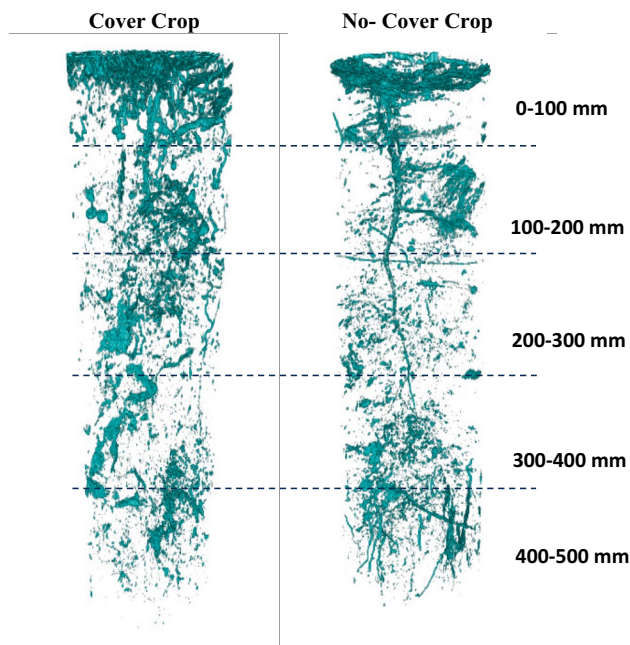


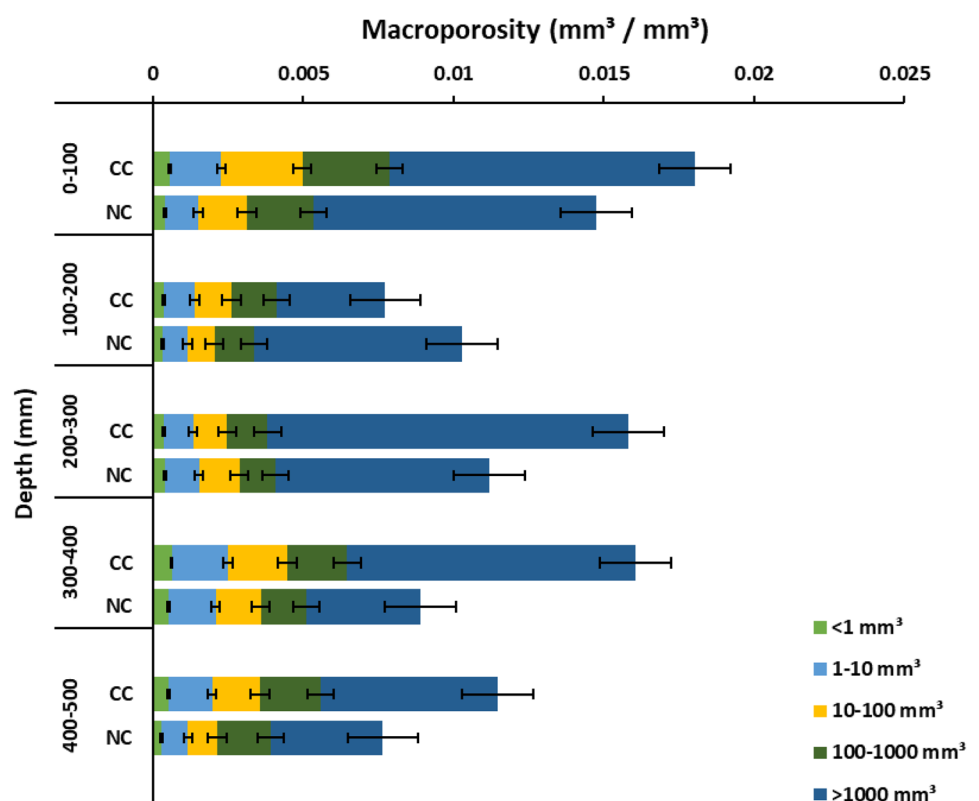
Fig. 1 3D Visualizations of macropore networks for soil columns under CC and NC treatments. The cyan color represents the soil pores

values. These findings are consistent with those of Haruna et al. (2018) and Zhang et al. (2022), who also reported greater proportions of larger pores under crop rows, with

most of the pore volume being occupied by root-induced biopores. Dal Ferro et al. (2013) suggested that the higher percentage of larger pores could be attributed to increased organic matter content and linked this to enhanced infiltration and drainage. These pores are more likely to act as the main conduits and main flow pathways for solute transport through the soil profile (Hu et al. 2019). There were no significant ($p < 0.05$) differences among the depth classes for porosity values of both the CC and NC treatments.

Macropore number density as a function of pore volume results indicates significant differences ($p < 0.05$) in treatments for the surface layer of the soil profile (0–100 mm) as shown in Fig. 3. When CC was compared to NC, it was observed that the CC treatment had a significantly ($p < 0.05$) greater number of macropores as compared to the NC treatment only for the surface layer (0–100 mm). Rankoth et al. (2022) also reported a greater number of pores for the cover crop treatment under a no-till management system for their surface depth class (0–60 mm). Significant differences ($p < 0.05$) among various depth classes were observed. For the CC treatments, the surface layer of 0–100 mm had significantly ($p < 0.05$) greater macropore number density as compared to the subsurface depth classes. Haruna et al. (2018) had similar results, and this was explained by the fact that as the root network dominated this soil depth, there may have been more accumulation of organic matter. For the NC treatment, macropore number density was

Fig. 2 Macroporosity distribution along the soil depth (mm) in CC and NC treatments under various volumetric sub classes ($< 1 \text{ mm}^3$, $1\text{--}10 \text{ mm}^3$, $10\text{--}100 \text{ mm}^3$, $100\text{--}1000 \text{ mm}^3$, and $> 1000 \text{ mm}^3$). Error bars represent the standard error of the mean for $n = 6$



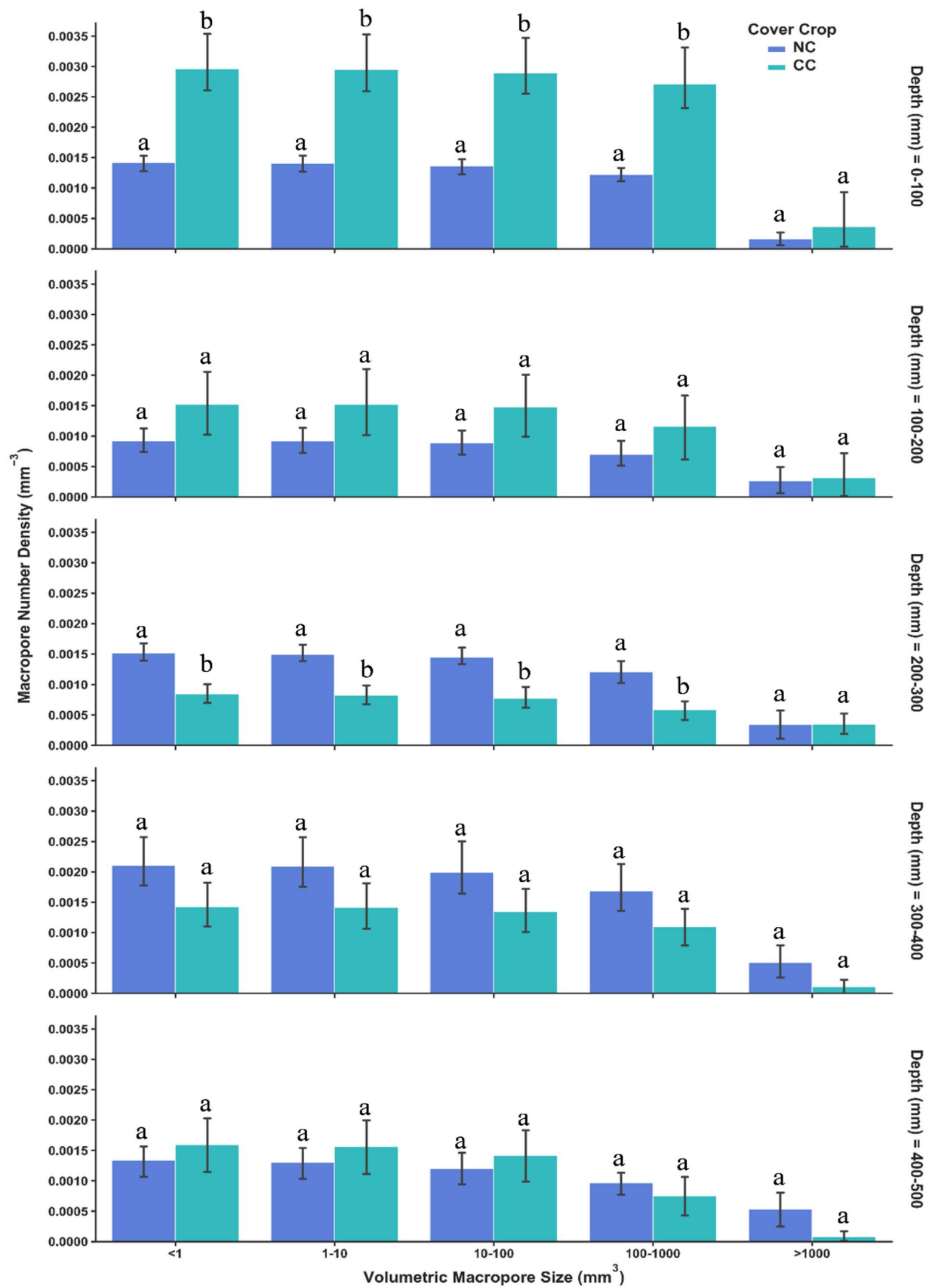


Fig. 3 Pore number density distribution along the soil depth (mm) for CC and NC treatments and volumetric subclasses. Error bars represent standard deviation ($n=6$). Within each depth class, different letters represent significantly ($p<0.05$) different mean macropore number density values

significantly ($p<0.05$) greater at the 200–300 mm depth. Overall, macropore number density was significantly ($p<0.05$) greater for CC when averaged across all the five depth zones with a value of 0.00279 mm^{-3} of ROI as compared to 0.00145 mm^{-3} for the NC treatment. This can be related to the porosity results, as porosity values were also highest for the CC treatment. Ghestem et al. (2011) reported that both live and dead components of the root systems tend to dissipate soil water pressure, and these root channels act as larger openings for rapid flow. No significant differences were observed for surface area density values between CC and NC treatments for 0–500 mm depth of soil profile.

4.3 Pore geometry descriptors

Contaminant transport in soils is greatly affected by the shape of pores present within the soil profile (Cercioglu et al. 2018). The morphology of pores defined by pore shape characteristics affects root development and pores along the roots and eventually influences the movement of solutes, water, and gases through the soil (Pires et al. 2019). Various soil management practices can directly influence the pore shape as there are changes in the way soil particles get arranged after the management practices have been implemented (Ogilvie et al. 2021).

Depth-wise distributions for circularity and macropore diameter are shown in Fig. 4a, b, respectively. The circularity values are greater for the CC treatment at the top 200 mm of the soil profile. Cercioglu et al. (2018) also found greater circularity values for their CC treatments and suggested that it could be due to greater root activity in this depth of the soil profile. The circularity values generally increased with depth. Macropore diameter represented the average pore size distribution for various treatments. There were significant ($p<0.05$) differences between treatments for some depth classes such as NC treatment had significantly ($p<0.05$) lower macropore diameter as compared to the CC treatment for the top 0–200 mm depth class. There were no significant differences ($p<0.05$) between treatments for subsequent depth classes.

The pores were further classified on the basis of their equivalent cylindrical diameter, i.e., very fine pores ($<1 \text{ mm}$), fine pores (1–2 mm), medium pores (2–5 mm), and coarse pores ($>5 \text{ mm}$). The circularity values were smaller; i.e., pores were elongated in shape for the coarser

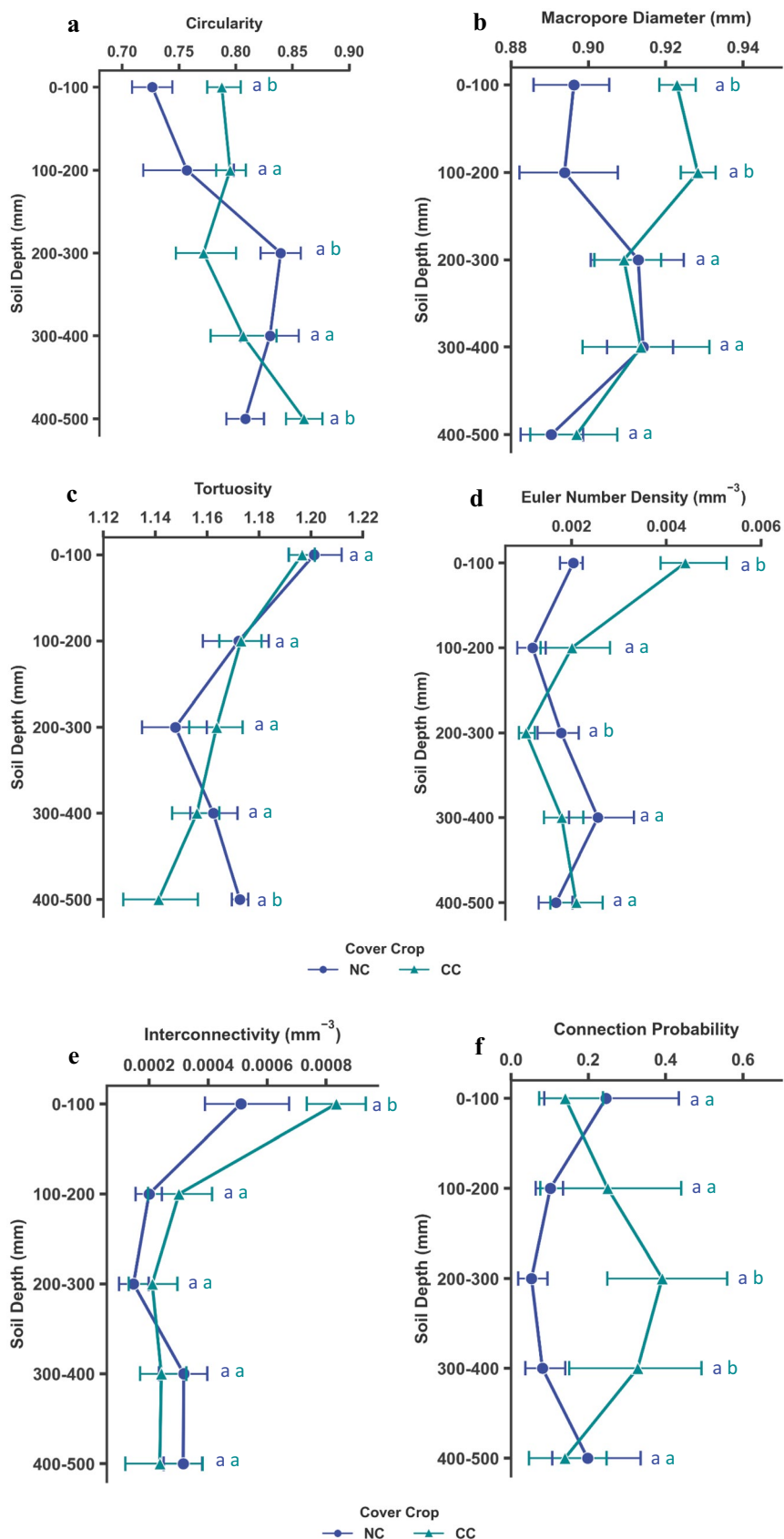
pores as compared to the finer pores for all depth classes and treatment classes. Yang et al. (2021) also found that smaller pores were more circular as compared to larger pores. Pires et al. (2019) suggested pores that were elongated in shape are created due to the presence of earthworms, and these elongated pores can cause enhanced transmission of water and gases (Pagliai et al. 2004). Zhao et al. (2017) observed a greater percentage of elongated pores in their study plots with restored vegetation as compared to their control plots.

Tortuosity is a complex geometrical parameter that defines the deflection of a pore network from its straight-line path (Shanti et al. 2014) and can affect the transport of solutes throughout the pore space (Moldrup et al. 2001). The flow rate through geometrically straight macropores is generally higher than tortuous macropores (Saravanathiiban et al. 2014). There were no significant differences between values of tortuosity for the CC and NC treatments. Dhaliwal and Kumar (2022) also did not find significant differences when a comparison was made between the tortuosity of their cover crop system and pasture land. However, significant differences were reported by Luo et al. (2010b) when comparisons were made between croplands and pasture land. In our study, the tortuosity values ranged from 1.11 to 1.22 (mean value of 1.165) under all treatment systems. The mean value of tortuosity found in our study was smaller compared to 1.25 reported by Katuwal et al. (2015) and 1.50 reported by Luo et al. (2010b) for agricultural soils under red fescue (*Festuca rubra* L.) and cropland, respectively. A lower tortuosity value suggests the presence of vertical pores. Crops with a tap root system, such as cotton in our case, can have continuous vertical macropores (Pagenkemper et al. 2013).

4.4 Connectivity metrics

Connectivity as defined by Renard and Allard (2013) is an indicator of spatially connected patterns that directly affect solute transport processes in a porous medium. These connected pathways create a percolating network (Koestel et al. 2018) for which various connectivity parameters can be used to generate relationships between pore continuity and permeability of the soil profile (Vogel 2000). Euler number density (χ) for the entire pore space for the defined volume of interest, which is a local connectivity measure, varied significantly ($p<0.05$) among the treatment classes. CC treatment had a significantly ($p<0.05$) higher value of χ for just the surface layer representing a greater number of isolated pore clusters in CC as compared to NC. A plausible reason could be the presence of pore clusters formed from finer cover crop root networks. Katuwal et al. (2015) reported similar results, as most of their macropore clusters were a part of isolated connections.

Fig. 4 Depth-wise distributions for **a** circularity, **b** macropore diameter (mm), **c** tortuosity, **d** Euler number density (mm^{-3}), **e** interconnectivity (mm^{-3}), **f** connection probability for CC and NC treatments. Error bars represent standard deviation ($n=6$). Within each graph, within each depth class, different letters represent significant differences ($p<0.05$)



However, the 200–300 mm depth class had significantly ($p < 0.05$) lower values of χ for CC treatments. This would most likely suggest the presence of a large proportion of local connections. Connection probability (Γ) or global continuity values account for the presence of a connected cluster throughout the ROI. As compared to Euler number density which is a local metric, connection probability is a global property (Herring et al. 2015); i.e., it does not depend on the presence of isolated pore clusters (Koestel et al. 2020) and is considered to be a reliable quantitative measure of connectivity (Renard and Allard 2013). The values of Γ did not vary significantly ($p < 0.05$) among the CC and NC treatments for the 0–200 mm depth but were significantly ($p < 0.05$) higher for CC treatment for the sub-surface depths (200–400 mm) as shown in Fig. 4f. These values symbolized the presence of pore clusters that were connected from top to bottom of the volume of interest (100 mm depth increments in our case). Xuan et al. (2022) have also summarized in their study that the presence of organic matter significantly improved pore properties, especially pore connectivity. The interconnectivity, defined as the number of nodes per unit volume of ROI (Budhathoki et al. 2022a), also varied significantly among treatments. Interconnectivity was significantly ($p < 0.05$) higher in CC treatment for the top 100 mm of the soil profile which signifies the presence of more nodes in the pore network of CC treatment. Node density depicts the connection of one pore to other pores. This value suggests the presence of an intricate pore network and is thus a clear indicator of soil pore connectivity (Xuan et al. 2022). The term interconnectivity or node density was defined as coordination number by Rankoth et al. (2022), and they have also reported greater values of node density in CC treatment as compared to NC treatment. This suggests the presence of more complex networks in the pores under CC treatment.

4.5 Relationship among CT-derived macropore properties and root characteristics

The Spearman correlation coefficients between various X-ray CT-derived macropore characteristics are shown in Table 3. Significant correlations were observed among several properties, which can help deduce some general relationships. These results indicate that larger pore volumes were associated with higher porosity values. These larger pore volumes, with relatively greater porosity values, were also positively correlated to the interconnectivity parameter. This might seem counter-intuitive, but it can be explained by the fact that larger pores were responsible for the highly interconnected macropore clusters. The macropore number density was also significantly positively correlated to the macropore circularity ($p < 0.05$ and Spearman correlation coefficient of 0.82) with the pore number density. This suggests that the CC treatment created a small number of elongated pores, and more number of circular pores. Pires et al. (2019) also suggested that there was a positive correlation between tri-axial pores and porosity and further indicated that the presence of these pores can significantly affect water infiltration. Connection probability was significantly ($p < 0.05$) positively correlated with macropore diameter, indicating larger macropores were associated with more connected macropore clusters. Cercioglu et al. (2018) also found significant correlations between CT-measured microporosity, pore circularity, area of the largest pore, and soil hydrologic parameters such as K_{sat} for fields with cover crops and bio-fuel crops. De Gryze et al. (2006) also found an increase in micro-cracks in aggregates close to organic matter area.

Table 4 shows that the soil macropore network properties correlate well with the root characteristics. Macropore network characteristics such as macroporosity, macropore volume, tortuosity, and connection probability showed significant

Table 3 Spearman correlation coefficients among different X-ray CT-derived soil pore characteristics across all treatment systems

	η	MND	σ	V	D	C	τ	I	Γ	χ
η		0.91***	0.95***	0.89***	0.32	0.58	-0.03	0.98***	0.83**	0.93***
MND			0.98***	0.63	-0.07	0.82**	-0.35	0.82**	0.57	1***
σ				0.7*	0.01	0.67	-0.14	0.91***	0.62	0.97
V					0.72*	0.27	0.21	0.92***	0.99***	0.69
D						-0.3	0.47	0.41	0.78**	0.01
C							-0.83**	0.39	0.28	0.81
τ								0.18	0.15	-0.34
I									0.85**	0.85**
Γ										0.63
χ										

η macroporosity, MND macropore number density, σ surface area density, V macropore volume, D macropore diameter, C circularity, τ tortuosity, I interconnectivity, Γ connection probability, χ Euler number

*, **, *** represent significance at 0.01, 0.05, and 0.1 level

Table 4 Spearman correlation coefficients among different X-ray CT-derived soil pore characteristics across all treatment systems and root characteristics

Macropore network characteristics	Root characteristics			
	Root length density	Root surface area density	Mean root diameter	Root volume
η	0.58***	0.5**	−0.09	0.39*
MND	−0.05	−0.11	−0.07	−0.14
σ	0.06	−0.16	−0.37	−0.26
V	0.62***	0.53**	−0.12	0.4*
D	0.4*	0.36	0.02	0.31
C	−0.39*	−0.63***	−0.48**	−0.71***
τ	0.46**	0.67***	0.38*	0.71***
I	0.31	0.41*	0.18	0.4*
F	0.64***	0.55**	−0.17	0.38*
χ	−0.05	−0.06	0.02	−0.06

η macroporosity, MND macropore number density, σ surface area density, V macropore volume, D macropore diameter, C circularity, τ tortuosity, I interconnectivity, F connection probability, χ Euler number

*, **, *** represents significance at 0.01, 0.05, and 0.1 level

positive correlations with the root characteristics. Correlations for macropore volume and root length density are consistent with the findings of Hu et al. (2019), who reported significant correlations between macropore networks and root characteristics for grasslands. Kochiieru et al. (2020) also found significant correlations among root length density and volume of macropores for grasslands, forest soils, and arable lands. We found that larger volumes of roots were associated with more tortuous macropore networks. However, macropore circularity was significantly negatively correlated with the root parameters. This suggested that roots with greater values of length, surface area, diameter, and volume were associated with irregularly shaped pores.

5 Conclusions

Results from this research study show that the X-ray CT-derived soil pore characteristics are greatly influenced by the cover crop management practices. Most significant differences were observed in the surface layer (0–100 mm). The volume descriptors (porosity and pore number density), geometry descriptors (circularity and macropore diameter), and connectivity descriptors (connection probability) were significantly greater in the CC treatment. These results were explained by the CC root-influenced pore properties. When correlations were performed, significant relations were found among the X-ray CT-scanned derived macropore properties and the WinRhizo-scanned root properties.

Future studies should improve the estimation accuracy for various pore characteristics by increasing the scanning resolution and quantifying contaminant loss in lateral and sub-surface flow pathways. Further research studies to quantify the influence of types of cover crops and their root properties on the macropore network characteristics will improve our understanding of the impact of cover cropping on soil physical properties.

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Data availability Data will be available upon request.

Declarations

Competing interests The authors declare no competing interests.

References

- Alaoui A, Goetz B (2008) Dye tracer and infiltration experiments to investigate macropore flow. *Geoderma* 144:279–286. <https://doi.org/10.1016/j.geoderma.2007.11.020>
- Anderson SH, Heinze DJ, Peyton RL (2015) Assessment of selected methods for estimating chemical transport parameters from computed tomographic imaging. *Procedia Comput Sci* 61:460–465. <https://doi.org/10.1016/j.procs.2015.09.188>
- Bölscher T, Koestel J, Etana A, Ulén B, Berglund K, Larsbo M (2021) Changes in pore networks and readily dispersible soil following structure liming of clay soils. *Geoderma* 390:114948. <https://doi.org/10.1016/j.geoderma.2021.114948>
- Bouyoucos GJ (1962) Hydrometer method improved for making particle size analyses of soils¹. *Agron J* 54:464–465. <https://doi.org/10.2134/agronj1962.00021962005400050028x>
- Budhathoki S, Lamba J, Srivastava P, Williams C, Arriaga F, Karthikeyan KG (2022a) Impact of land use and tillage practice on soil macropore characteristics inferred from X-ray computed tomography. *CATENA* 210:105886. <https://doi.org/10.1016/j.catena.2021.105886>
- Budhathoki S, Lamba J, Srivastava P, Malhotra K, Way TR, Katuwal S (2022b) Using X-ray computed tomography to quantify variability in soil macropore characteristics in pastures. *Soil Tillage Res* 215:105194. <https://doi.org/10.1016/j.still.2021.105194>
- Budhathoki S, Lamba J, Srivastava P, Malhotra K, Way TR, Katuwal S (2022c) Temporal and spatial variability in 3D soil macropore characteristics determined using X-ray computed tomography. *J Soils Sediments* 22:1263–1277. <https://doi.org/10.1007/s11368-022-03150-x>
- Busari MA, Kukal SS, Kaur A, Bhatt R, Dulazi AA (2015) Conservation tillage impacts on soil, crop and the environment. *Int Soil Water Conserv Res* 3:119–129. <https://doi.org/10.1016/j.iswcr.2015.05.002>
- Buttle JM, McDonald DJ (2002) Coupled vertical and lateral preferential flow on a forested slope. *Water Resour Res* 38:18–1–18–16. <https://doi.org/10.1029/2001WR000773>

- Çerçioğlu M (2020) Changes in soil hydro-physical properties by cover crops relative to tillage management. *Eurasian Soil Sci* 53:1446–1454. <https://doi.org/10.1134/S1064229320100051>
- Çerçioğlu M, Anderson SH, Udawatta RP, Alagele S (2019) Effect of cover crop management on soil hydraulic properties. *Geoderma* 343:247–253. <https://doi.org/10.1016/j.geoderma.2019.02.027>
- Cercioglu M, Anderson SH, Udawatta RP, Haruna SI (2018) Effects of cover crop and biofuel crop management on computed tomography-measured pore parameters. *Geoderma* 319:80–88. <https://doi.org/10.1016/j.geoderma.2018.01.005>
- Chakraborty P, Singh N, Bansal S, Sekaran U, Sexton P, Bly A, Anderson SH, Kumar S (2022) Does the duration of no-till implementation influence depth distribution of soil organic carbon, hydro-physical properties, and computed tomography-derived macropore characteristics? *Soil Tillage Res* 222:105426. <https://doi.org/10.1016/j.still.2022.105426>
- Chu Z, Cheng Y, Zhang Q, Zhou H, Dai Y, Shi Y, Gregori G, Rosenfeld PJ, Wang RK (2020) Quantification of choriocapillaris with Phansalkar local thresholding: pitfalls to avoid. *Am J Ophthalmol* 213:161–176. <https://doi.org/10.1016/j.ajo.2020.02.003>
- Dal Ferro N, Charrier P, Morari F (2013) Dual-scale micro-CT assessment of soil structure in a long-term fertilization experiment. *Geoderma* 204–205:84–93. <https://doi.org/10.1016/j.geoderma.2013.04.012>
- Dal Ferro N, Sartori L, Simonetti G, Berti A, Morari F (2014) Soil macro- and microstructure as affected by different tillage systems and their effects on maize root growth. *Soil Tillage Res* 140:55–65. <https://doi.org/10.1016/j.still.2014.02.003>
- De Gryze S, Jassogne L, Six J, Bossuyt H, Wevers M, Merckx R (2006) Pore structure changes during decomposition of fresh residue: X-ray tomography analyses. *Geoderma* 134:82–96. <https://doi.org/10.1016/j.geoderma.2005.09.002>
- Dhaliwal JK, Kumar S (2022) 3D-visualization and quantification of soil porous structure using X-ray micro-tomography scanning under native pasture and crop-livestock systems. *Soil Tillage Res* 218:105305. <https://doi.org/10.1016/j.still.2021.105305>
- Doube M, Kłosowski MM, Arganda-Carreras I, Cordelières FP, Dougherty RP, Jackson JS, Schmid B, Hutchinson JR, Shefelbine SJ (2010) BoneJ: free and extensible bone image analysis in ImageJ. *Bone* 47:1076–1079. <https://doi.org/10.1016/j.bone.2010.08.023>
- Filipović V, Defterdarović J, Šimunek J, Filipović L, Ondrašek G, Romić D, Bogunović I, Mustać I, Čurić J, Kodešová R (2020) Estimation of vineyard soil structure and preferential flow using dye tracer, X-ray tomography, and numerical simulations. *Geoderma* 380:114699. <https://doi.org/10.1016/j.geoderma.2020.114699>
- Gantzer CJ, Anderson SH (2002) Computed tomographic measurement of macroporosity in chisel-disk and no-tillage seedbeds. *Soil Tillage Res* 64:101–111. [https://doi.org/10.1016/S0167-1987\(01\)00248-3](https://doi.org/10.1016/S0167-1987(01)00248-3)
- Garbout A, Munkholm LJ, Hansen SB (2013) Tillage effects on topsoil structural quality assessed using X-ray CT, soil cores and visual soil evaluation. *Soil Tillage Res* 128:104–109. <https://doi.org/10.1016/j.still.2012.11.003>
- Ghestem M, Sidle RC, Stokes A (2011) The influence of plant root systems on subsurface flow: implications for slope stability. *BioScience* 61:869–879. <https://doi.org/10.1525/bio.2011.61.11.6>
- Haruna SI, Anderson SH, Nkongolo NV, Zaibon S (2018) cerci. *Pedosphere* 28:430–442. [https://doi.org/10.1016/S1002-0160\(17\)60387-4](https://doi.org/10.1016/S1002-0160(17)60387-4)
- Haruna SI, Anderson SH, Udawatta RP, Gantzer CJ, Phillips NC, Cui S, Gao Y (2020) Improving soil physical properties through the use of cover crops: a review. *Agrosystems Geosci Environ*. <https://doi.org/10.1002/agg2.20105>
- Helliwell JR, Sturrock CJ, Miller AJ, Whalley WR, Mooney SJ (2019) The role of plant species and soil condition in the structural development of the rhizosphere. *Plant Cell Environ* 42:1974–1986. <https://doi.org/10.1111/pce.13529>
- Hellner Q, Koestel J, Ulén B, Larsbo M (2018) Effects of tillage and liming on macropore networks derived from X-ray tomography images of a silty clay soil. *Soil Use Manag* 34:197–205. <https://doi.org/10.1111/sum.12418>
- Herring AL, Andersson L, Schlüter S, Sheppard A, Wildenschild D (2015) Efficiently engineering pore-scale processes: the role of force dominance and topology during nonwetting phase trapping in porous media. *Adv Water Resour* 79:91–102. <https://doi.org/10.1016/j.advwatres.2015.02.005>
- Hu X, Li X-Y, Wang P, Liu Y, Wu XC, Li ZC, Zhao YD, Cheng YQ, Guo LL, Lyu YL, Liu LY (2019) Influence of enclosure on CT-measured soil macropores and root architecture in a shrub-encroached grassland in northern China. *Soil Tillage Res* 187:21–30. <https://doi.org/10.1016/j.still.2018.10.020>
- Jarvis N, Larsbo M, Koestel J (2017) Connectivity and percolation of structural pore networks in a cultivated silt loam soil quantified by X-ray tomography. *Geoderma* 287:71–79. <https://doi.org/10.1016/j.geoderma.2016.06.026>
- Kaspar TC, Singer JW (2011) The Use of Cover Crops to Manage Soil. In: *Soil Management: Building a Stable Base for Agriculture*. John Wiley & Sons, Ltd, pp 321–337. <https://doi.org/10.2136/2011.soilmanagement.c21>
- Katuwal S, Norgaard T, Moldrup P, Lamandé M, Wildenschild D, De Jonge L (2015) Linking air and water transport in intact soils to macropore characteristics inferred from X-ray computed tomography. *Geoderma* 237–238:9–20. <https://doi.org/10.1016/j.geoderma.2014.08.006>
- Ketema H, Yimer F (2014) Soil property variation under agroforestry based conservation tillage and maize based conventional tillage in Southern Ethiopia. *Soil Tillage Res* 141:25–31. <https://doi.org/10.1016/j.still.2014.03.011>
- Kochiieru M, Lamorski K, Feiza V, Feizienė D, Volungevičius J (2018) The effect of soil macroporosity, temperature and water content on CO₂ efflux in the soils of different genesis and land management. *Zemdirb-Agric* 105:291–298. <https://doi.org/10.13080/z-a.2018.105.037>
- Kochiieru M, Lamorski K, Feiza V, Feizienė D, Volungevičius J (2020) Quantification of the relationship between root parameters and soil macropore parameters under different land use systems in Retisol. *Int Agrophysics* 34:301–308. <https://doi.org/10.31545/intagr/123266>
- Koebnick N, Daly KR, Keyes SD, Bengough AG, Brown LK, Cooper LJ, George TS, Hallett PD, Naveed M, Raffan A, Roose T (2019) Imaging microstructure of the barley rhizosphere: particle packing and root hair influences. *New Phytol* 221:1878–1889. <https://doi.org/10.1111/nph.15516>
- Koestel J (2018) SoilJ: an ImageJ plugin for the semiautomatic processing of three-dimensional X-ray images of soils. *Vadose Zone J* 17:170062. <https://doi.org/10.2136/vzj2017.03.0062>
- Koestel J, Dathe A, Skaggs TH, Klakegg O, Ahmad MA, Babko M, Giménez D, Farkas C, Nemes A, Jarvis N (2018) Estimating the permeability of naturally structured soil from percolation theory and pore space characteristics imaged by X-ray. *Water Resour Res* 54:9255–9263. <https://doi.org/10.1029/2018WR023609>
- Koestel J, Fukumasu J, Garland G, Larsbo M, Nimblad Svensson D (2021) Approaches to delineate aggregates in intact soil using X-ray imaging. *Geoderma* 402:115360. <https://doi.org/10.1016/j.geoderma.2021.115360>
- Koestel J, Larsbo M, Jarvis N (2020) Scale and REV analyses for porosity and pore connectivity measures in undisturbed soil. *Geoderma* 366:114206. <https://doi.org/10.1016/j.geoderma.2020.114206>

- Kornecki TS, Prior S, Torbert H (2016) Effects of a Custom Cover Crop Residue Manager in a No-Till Cotton System. *Appl Eng Agric* 32:333–340. <https://doi.org/10.13031/aea.32.11567>
- Kornecki TS (2018) Effect of different rolling/crimping directions and row cleaners for cereal rye cover crop on cotton growth in a conservation system. *Trans ASABE* 61:1845–1855. <https://doi.org/10.13031/trans.12635>
- Kumar H, Srivastava P, Ortiz BV, Morata H, Takhellambam BS, Lamba J, Bondesan L (2021) Field-scale spatial and temporal soil water variability in irrigated croplands. *Trans ASABE* 64:1277–1294. <https://doi.org/10.13031/trans.14335>
- Larsbo M, Koestel J, Jarvis N (2014) Relations between macropore network characteristics and the degree of preferential solute transport. *Hydrol Earth Syst Sci* 18:5255–5269. <https://doi.org/10.5194/hess-18-5255-2014>
- Li P, Shao S (2020) Can X-ray computed tomography (CT) be used to determine the pore-size distribution of intact loess? *Environ Earth Sci* 79:29. <https://doi.org/10.1007/s12665-019-8777-z>
- Li T, Shao M, Jia Y, Jia X, Huang Y (2018) Using the X-ray computed tomography method to predict the saturated hydraulic conductivity of the upper root zone in the Loess Plateau in China. *Soil Sci Soc Am J* 82:1085–1092. <https://doi.org/10.2136/sssaj2017.08.0268>
- Liu Z, Liu F, Ma F, Wang M, Bai X, Zheng Y, Yin H, Zhang G (2016) Collapsibility, composition, and microstructure of loess in China. *Can Geotech J* 53:673–686. <https://doi.org/10.1139/cgj-2015-0285>
- Lucas M, Vetterlein D, Vogel H, Schlüter S (2021) Revealing pore connectivity across scales and resolutions with X-ray CT. *Eur J Soil Sci* 72:546–560. <https://doi.org/10.1111/ejss.12961>
- Luo L, Lin H, Li S (2010a) Quantification of 3-D soil macropore networks in different soil types and land uses using computed tomography. *J Hydrol* 393:53–64. <https://doi.org/10.1016/j.jhydrol.2010.03.031>
- Luo L, Lin H, Schmidt J (2010b) Quantitative relationships between soil macropore characteristics and preferential flow and transport. *Soil Sci Soc Am J* 74:1929–1937. <https://doi.org/10.2136/sssaj2010.0062>
- Mangalassery S, Sjögersten S, Sparkes DL, Sturrock CJ, Craigon J, Mooney SJ (2014) To what extent can zero tillage lead to a reduction in greenhouse gas emissions from temperate soils? *Sci Rep* 4:4586. <https://doi.org/10.1038/srep04586>
- Mao N, Li W, Wei X, Shao M (2020) Soil moisture- and texture-dependent effects of soil deposition on evaporation and carbon emission. *Soil Tillage Res* 204:104703. <https://doi.org/10.1016/j.still.2020.104703>
- Miranda-Vélez JF, Leuther F, Köhne JM, Munkholm LJ, Vogeler I (2023) Effects of freeze-thaw cycles on soil structure under different tillage and plant cover management practices. *Soil Tillage Res* 225:105540. <https://doi.org/10.1016/j.still.2022.105540>
- Moldrup P, Olesen T, Komatsu T, Schjønning P, Rolston DE (2001) Tortuosity, diffusivity, and permeability in the soil liquid and gaseous phases. *Soil Sci Soc Am J* 65:613–623. <https://doi.org/10.2136/sssaj2001.653613x>
- Müller K, Katuwal S, Young I, McLeod M, Moldrup P, De Jonge LW, Clothier B (2018) Characterising and linking X-ray CT derived macroporosity parameters to infiltration in soils with contrasting structures. *Geoderma* 313:82–91. <https://doi.org/10.1016/j.geoderma.2017.10.020>
- Munkholm LJ, Heck RJ, Deen B (2013) Long-term rotation and tillage effects on soil structure and crop yield. *Soil Tillage Res* 127:85–91. <https://doi.org/10.1016/j.still.2012.02.007>
- Muñoz FJ, San José Martínez F, Caniego FJ (2014) Fractal parameters of pore space from CT images of soils under contrasting management practices. *Fractals* 22:1440011. <https://doi.org/10.1142/S0218348X14400118>
- Nichele L, Persichetti V, Lucidi M, Cincotti G (2020) Quantitative evaluation of ImageJ thresholding algorithms for microbial cell counting. *OSA Contin* 3:1417. <https://doi.org/10.1364/OSAC.393971>
- Noguchi S, Tsuboyama Y, Sidle RC, Hosoda I (1999) Morphological characteristics of macropores and the distribution of preferential flow pathways in a forested slope segment. *Soil Sci Soc Am J* 63:1413–1423. <https://doi.org/10.2136/sssaj1999.6351413x>
- Ogilvie CM, Ashiq W, Vasava HB, Biswas A (2021) Quantifying root-soil interactions in cover crop systems: a review. *Agriculture* 11:218. <https://doi.org/10.3390/agriculture11030218>
- Pagenkemper SK, Peth S, Puschmann DU, Horn R (2013) Effects of Root-Induced Biopores on Pore Space Architecture Investigated with Industrial X-Ray Computed Tomography. In: *Soil–Water–Root Processes: Advances in Tomography and Imaging*. John Wiley & Sons, Ltd, pp 69–96. <https://doi.org/10.2136/sssaspecpub61.c4>
- Pagliai M, Vignozzi N, Pellegrini S (2004) Soil structure and the effect of management practices. *Soil Tillage Res* 79:131–143. <https://doi.org/10.1016/j.still.2004.07.002>
- Peth S, Horn R, Beckmann F, Donath T, Fischer J, Smucker AJM (2008) Three-dimensional quantification of intra-aggregate pore-space features using synchrotron-radiation-based microtomography. *Soil Sci Soc Am J* 72:897–907. <https://doi.org/10.2136/sssaj2007.0130>
- Phalempin M, Lippold E, Vetterlein D, Schlüter S (2021) Soil texture and structure heterogeneity predominantly governs bulk density gradients around roots. *Vadose Zone J*. <https://doi.org/10.1002/vzj.20147>
- Phansalkar N, More S, Sabale A, Joshi M (2011) Adaptive local thresholding for detection of nuclei in diversity stained cytology images. In: *2011 International Conference on Communications and Signal Processing*. IEEE, Kerala, India, pp 218–220. <https://doi.org/10.1109/ICCSP.2011.5739305>
- Pires LF, Roque WL, Rosa JA, Mooney SJ (2019) 3D analysis of the soil porous architecture under long term contrasting management systems by X-ray computed tomography. *Soil Tillage Res* 191:197–206. <https://doi.org/10.1016/j.still.2019.02.018>
- Prior SA, Runion GB, Torbert HA, Erbach DC (2004) A hydraulic coring system for soil-root studies. *Agron J* 96:1202–1205. <https://doi.org/10.2134/agronj2004.1202>
- Pulido-Moncada M, Katuwal S, Ren L, Cornelis W, Munkholm L (2020) Impact of potential bio-subsoilers on pore network of a severely compacted subsoil. *Geoderma* 363:114154. <https://doi.org/10.1016/j.geoderma.2019.114154>
- Qiu L, Chen X, Pan J (2013) In situ measurement of soil macropores by dye tracing and image analysis. In: *2013 Second International Conference on Agro-Geoinformatics (Agro-Geoinformatics)*. IEEE, Fairfax, VA, USA, pp 13–17. <https://doi.org/10.1109/Argo-Geoinformatics.2013.6621870>
- Rabot E, Wiesmeier M, Schlüter S, Vogel H-J (2018) Soil structure as an indicator of soil functions: a review. *Geoderma* 314:122–137. <https://doi.org/10.1016/j.geoderma.2017.11.009>
- Rankoth LM, Udawatta RP, Gantzer CJ, Anderson SH (2022) Cover crop effects on μ CT-measured geometrical pore characteristics. *Agrosyst Geosci Environ*. <https://doi.org/10.1002/agg2.20284>
- Renard P, Allard D (2013) Connectivity metrics for subsurface flow and transport. *Adv Water Resour* 51:168–196. <https://doi.org/10.1016/j.advwatres.2011.12.001>
- Saravanathiiban DS, Kutay ME, Khire MV (2014) Effect of macropore tortuosity and morphology on preferential flow through saturated soil: a lattice Boltzmann study. *Comput Geotech* 59:44–53. <https://doi.org/10.1016/j.compgeo.2014.02.006>
- Schaap MG, Leij FJ, van Genuchten MTh (2001) rosetta: a computer program for estimating soil hydraulic parameters with hierarchical pedotransfer functions. *J Hydrol* 251:163–176. [https://doi.org/10.1016/S0022-1694\(01\)00466-8](https://doi.org/10.1016/S0022-1694(01)00466-8)
- Schindelin J, Arganda-Carreras I, Frise E, Kaynig V, Longair M, Pietzsch T, Preibisch S, Rueden C, Saalfeld S, Schmid B, Tinevez J, White DJ, Hartenstein V, Eliceiri K, Tomancak P, Cardona A (2012) Fiji:

- an open-source platform for biological-image analysis. *Nat Methods* 9:676–682. <https://doi.org/10.1038/nmeth.2019>
- Schlüter S, Eickhorst T, Mueller CW (2019) Correlative imaging reveals holistic view of soil microenvironments. *Environ Sci Technol* 53:829–837. <https://doi.org/10.1021/acs.est.8b05245>
- Schulte EE, Hopkins BG (1996) Estimation of soil organic matter by weight loss-on-ignition. in: soil organic matter: analysis and interpretation. John Wiley & Sons, Ltd, pp 21–31. <https://doi.org/10.2136/sssaspecpub46.c3>
- Shafait F, Keyers D, Breuel TM (2008) Efficient implementation of local adaptive thresholding techniques using integral images. In: Yanikoglu BA, Berkner K (eds). San Jose, CA, pp 681510–681510–6. <https://doi.org/10.1117/12.767755>
- Shanti NO, Chan VWL, Stock SR, De Carlo F, Thornton K, Faber KT (2014) X-ray micro-computed tomography and tortuosity calculations of percolating pore networks. *Acta Mater* 71:126–135. <https://doi.org/10.1016/j.actamat.2014.03.003>
- Singh J, Singh N, Kumar S (2020) X-ray computed tomography–measured soil pore parameters as influenced by crop rotations and cover crops. *Soil Sci Soc Am J* 84:1267–1279. <https://doi.org/10.1002/saj2.20105>
- Singh N, Kumar S, Udawatta RP, Anderson SH, De Jonge LW, Katuwal K (2021) X-ray micro-computed tomography characterized soil pore network as influenced by long-term application of manure and fertilizer. *Geoderma* 385:114872. <https://doi.org/10.1016/j.geoderma.2020.114872>
- Taina IA, Heck RJ, Elliot TR (2008) Application of X-ray computed tomography to soil science: a literature review. *Can J Soil Sci* 88:1–19. <https://doi.org/10.4141/CJSS06027>
- Terribile F, Langella G, Mileti FA, Minieri L, Vingiani S, Jon Mooney S, Young LM, Heck RJ, Peth S (2022) Soil Sampling and Preparation for X-ray Imaging. In: Jon Mooney S, Young IM, Heck RJ, Peth S (eds) X-ray imaging of the soil porous architecture. Springer International Publishing, Cham, pp 19–38. https://doi.org/10.1007/978-3-031-12176-0_3
- Tracy SR, Daly KR, Sturrock CJ, Crout NMJ, Mooney SJ, Roose T (2015) Three-dimensional quantification of soil hydraulic properties using X-ray computed tomography and image-based modeling. *Water Resour Res* 51:1006–1022. <https://doi.org/10.1002/2014WR016020>
- Vogel HJ (2000) A numerical experiment on pore size, pore connectivity, water retention, permeability, and solute transport using network models: pore structure and hydraulic properties. *Eur J Soil Sci* 51:99–105. <https://doi.org/10.1046/j.1365-2389.2000.00275.x>
- Wang K, Zhang R (2011) Heterogeneous soil water flow and macropores described with combined tracers of dye and iodine. *J Hydrol* 397:105–117. <https://doi.org/10.1016/j.jhydrol.2010.11.037>
- Wildenschild D, Vaz CMP, Rivers ML, Rikard D, Christensen BSB (2002) Using X-ray computed tomography in hydrology: systems, resolutions, and limitations. *J Hydrol* 267:285–297. [https://doi.org/10.1016/S0022-1694\(02\)00157-9](https://doi.org/10.1016/S0022-1694(02)00157-9)
- Wolschick NH, Bagio B, Andrade AP, Rauber LP, Bertol I, Borg H (2021) The effect of soil management on pore size distribution and water infiltration. *Braz Arch Biol Technol* 64:e21190602. <https://doi.org/10.1590/1678-4324-2021190602>
- Xuan K, Li X, Yu X, Jiang Y, Ji J, Jia R, Wang C, Liu J (2022) Effects of different organic amendments on soil pore structure acquired by three-dimensional investigation. *Eur J Soil Sci*. <https://doi.org/10.1111/ejss.13264>
- Yang B, Kang Y, Lu X, You L, Zhang H, Chen Z (2020) Experimental investigation of the pore shape factor in fluid imbibition model--taking the Longmaxi shale in Sichuan Basin as examples. *J Pet Sci Eng* 193:107327. <https://doi.org/10.1016/j.petrol.2020.107327>
- Yang Y, Wu J, Zhao S, Mao Y, Zhang J, Pan X, He F, Ploeg M (2021) Impact of long-term sub-soiling tillage on soil porosity and soil physical properties in the soil profile. *Land Degrad Dev* 32:2892–2905. <https://doi.org/10.1002/ldr.3874>
- Zhang J, Xu Z, Li F, Hou RJ, Ren Z (2017) Quantification of 3D macropore networks in forest soils in Touzhai valley (Yunnan, China) using X-ray computed tomography and image analysis. *J Mt Sci* 14:474–491. <https://doi.org/10.1007/s11629-016-4150-9>
- Zhang R, Wienhold BJ (2002) The effect of soil moisture on mineral nitrogen, soil electrical conductivity, and pH. *Nutr Cycl Agroecosystems* 63:251–254. <https://doi.org/10.1023/A:1021115227884>
- Zhang Y, Schaap MG (2017) Weighted recalibration of the Rosetta pedotransfer model with improved estimates of hydraulic parameter distributions and summary statistics (Rosetta3). *J Hydrol* 547:39–53. <https://doi.org/10.1016/j.jhydrol.2017.01.00>
- Zhang Y, Jia A, Zhao W et al (2022) Soil macroporosity and water flow in the root zone of oases in hyper-arid regions. *Eur J Soil Sci* 73. <https://doi.org/10.1111/ejss.13235>
- Zhang Z, Liu K, Zhou H, Lin H, Li D, Peng X (2018) Three dimensional characteristics of biopores and non-biopores in the sub-soil respond differently to land use and fertilization. *Plant Soil* 428:453–467. <https://doi.org/10.1007/s11104-018-3689-3>
- Zhao D, Xu M, Liu G, Yao X, Tuo D, Zhang R, Xiao T, Peng G (2017) Quantification of soil aggregate microstructure on abandoned cropland during vegetative succession using synchrotron radiation-based micro-computed tomography. *Soil Tillage Res* 165:239–246. <https://doi.org/10.1016/j.still.2016.08.007>

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